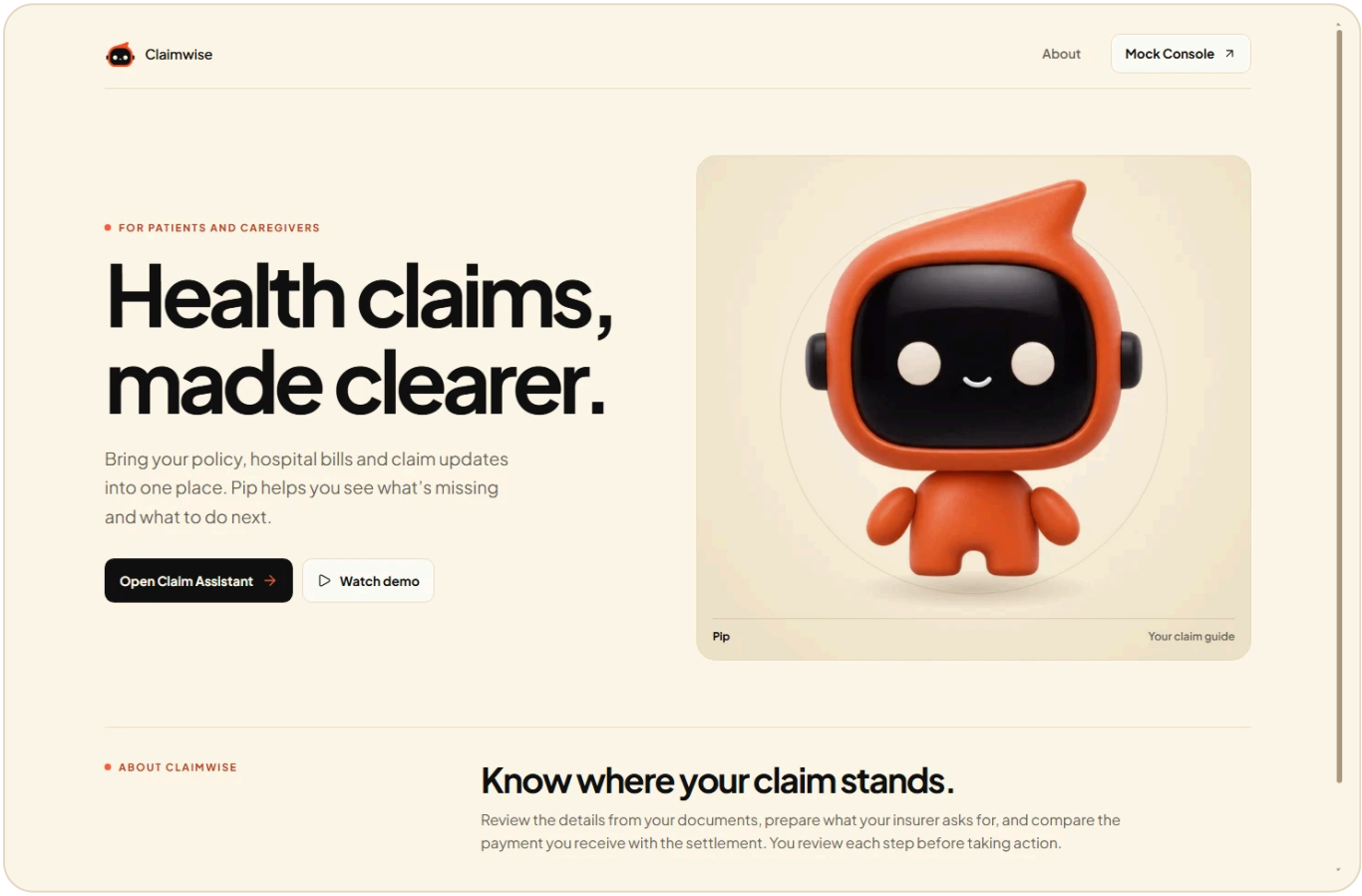
Claimwise

A health insurance claim lifecycle agent for patients, caregivers, and households.



Synthetic-data prototype

Product design assignment 03 · Revised brief, 1 November 2026

Repository snapshot 0d32dd8 with the submission documentation and diagram updates · 7 November 2026

Project repository · Product walkthrough

Submission map

The PDF consolidates the design. The repository links open the editable graphs, implementation, and detailed evidence.

Brief artifact	In this PDF	Repository evidence
Product brief and research	3–4	Brief · Evidence register
Personas, lifecycle and branches	4–7	Personas · Lifecycle map · Branch matrix
Five technical graphs	8–12	Graph index, legend and editable sources
Technical sheet and payloads	13–15	Technical reference · Function catalogue
UX, validation and delivery	16, 18–19	UX sketches · Validation · Phases
Walkthrough and three journeys	16–17	Written walkthrough and fixture sequence

Evidence boundaries. The diagrams describe a synthetic prototype. The cited validation is the recorded 6 November run. This submission uses a written walkthrough and live app captures. A recorded walkthrough remains a separate artifact.

Assignment traceability · Decisions and assumptions

One claim, from evidence to outcome

Claimwise gives the claimant a case file for the work currently spread across folders, insurer inboxes, reminders, and payment spreadsheets.

Capability	Working prototype	Proposed first release
Prepare	Read evidence, confirm policy facts, check gaps and export a pack.	Execute organization and checks. Ask for corrections when evidence is uncertain.
Submit and track	Send an approved pack through a mock TPA grant or record a manual handoff.	Coordinate claimant submission until live access and receipts are verified.
Requests and decisions	Flag repeats, review deductions, draft responses and track follow-up.	Execute comparison and drafting. The claimant verifies and sends.
Payment and disputes	Match reimbursement evidence, support a balance round and prepare grievance or Ombudsman drafts.	Reconcile and flag uncertainty. Humans file, review eligibility and confirm ambiguous credits.

User control

Confirm the patient and policy, choose the claim route or cover, correct extracted facts, review the exact pack, and approve consequential actions. Pip shows the next step and the proof still needed.

Decisions and open scope

D-01 to D-08 record the public submission package, one prototype deliverable, dependency-ordered review, private research inputs, synthetic data, parallel documentation and build, the approved prototype stack, and Vercel preparation. OQ-01 to OQ-07 keep jurisdiction, audience, route, replacement category, specialist boundary, release autonomy and evidence gaps open.

> **Scope boundary.** India is the research and demo context. Reimbursement-first with cashless coordination is a proposal. Medical necessity and legal representation require a specialist.

Product brief · Decision log

Different claimants, recurring gaps

The register distinguishes firsthand reports, documented facts, repeated patterns, and hypotheses. The evidence is qualitative.

Situation	Observed signal	Design response
Patient or parent · E-01, E-13	Cashless delay split one event into two routes. Another portal showed settled without a bank credit.	Keep one case across routes and require route-specific payment evidence.
Caregiver · E-06, E-07	Repeated evidence requests across handlers delayed claims, even after re-upload or escalation.	Keep a dated submission record and link the request to the evidence already sent.
Household with two covers · E-02, E-03	The second cover depends on originals or the first settlement summary.	Ask the claimant to choose primary cover and carry the first decision into the balance round.
Dispute and follow-up · E-08, E-09	Deductions lacked a clear rationale in one account. In another, an award paid the claim while reinstatement remained unresolved.	Separate deduction review, filing, payment and continuing obligations.

Alternatives

Manual email and file tracking spans the lifecycle with substantial claimant effort. Hospital desks help with cashless coordination. Agents and brokers vary in follow-through. Forums and general chatbots help with individual questions. A single drafting or reminder feature would offer little improvement over these options.

Limits and contrary evidence

E-01 reports a subsequent reimbursement without trouble. Forum reports over-represent problems, and some outcomes are unknown. The exact bank-credit mismatch and mid-case authority withdrawal remain evidence gaps. Treat them as design scenarios. No prevalence, market size or willingness-to-pay estimate follows from these reports.

Full situation register · Personas and situations · Alternatives · Sources and provenance

Prepare, submit, and wait

The same case thread carries authority, evidence, approvals and receipts between turns. Each checkpoint has an actor and observable evidence.

Checkpoint and intent	Agent work and waiting condition	Proof and next actor
01 · Before submission **Triggered by an upload or a new case. The claimant wants a complete, correct pack.**	O1 reads policy and medical evidence. O2 checks the patient and authority. O3 pauses for correction. C1 checks dates, names, bill amounts and required documents. Missing or contradictory evidence keeps the case in review.	Confirmed policy and role, required evidence, no unresolved blockers, and approval of the exact pack hash. The claimant confirms and selects the route.
02 · During submission **Triggered by pack approval. The claimant wants the right evidence accepted once.**	X1 checks approval and caregiver scope. The connected mock path sends through the gateway. The manual path produces a pack and asks for the receipt or submission date. An ambiguous write is marked unconfirmed.	Authoritative provider reference or acknowledgement, distinct from user-reported manual submission. The claimant or mock payer acts next. Silence triggers lookup before retry.
03 · While waiting **Triggered by submission, a signed update, reopening the case, or a user update.**	M1 reconciles uncertain submissions, reads authorized mock status and checks reminders. C1 classifies requests and links renamed repeats to evidence. Await missing documents or approval of the reply.	Acknowledged response or payer record resolves the request. The user supplies missing evidence or manually follows up. Silence surfaces a dated reminder and leaves the claim open.

Write lifecycle. Proposed → approved → attempted → confirmed. Cancellation, changed evidence, revocation and uncertain delivery have separate exits. A draft or HTTP request alone does not establish the external outcome.

Lifecycle map · Lifecycle states · Submission and recovery · Journey evidence

Decide, reconcile, and close

A decision can create further work. Payment, grievance compliance and future coverage may have different outcomes.

Checkpoint and intent	Agent work and waiting condition	Proof and next actor
04 · At a decision **Triggered by a provider update or a letter. The user wants to know what was allowed and why.**	C1 reads the decision and flags partial, refused or unclear items and missing clauses. A revised decision updates the affected outcome. Wait for clarification or supporting evidence if reasons are incomplete.	Dated decision, line amounts and supporting terms. User accepts, corrects evidence or asks the payer for reasons.
05 · After a decision **Triggered by settlement, a bank statement or a dispute. The user wants the money and a resolution.**	P1 compares reimbursement credit by available evidence, including UTR where present. X1 prepares grievance or Ombudsman drafts. A human approves and files. A revised award can leave a top-up due.	Matched bank credit, authoritative cashless settlement evidence, or recorded dispute outcome and compliance. Bank, payer or claimant acts next on mismatch or silence.
06 · After resolution **Triggered by verified outcome or a later change. The user wants a record and a safe next step.**	Review receipt and remaining obligations. An earlier policy round can support a balance claim. Changed evidence reopens the affected check. User may delete case uploads or the session. Production retention and reuse permissions need approval.	Documented closure evidence. A late credit, revised decision, unresolved obligation or second cover can reopen work with its own next actor.

Completion by object

Submitted item → provider acknowledgement. Answered request → accepted response or payer state. Decision → recorded outcome and applicable reasons. Payment → matched evidence on the correct route. Dispute → outcome plus evidence of compliance when due.

Known demo gap. The walkthrough records that the cashless workspace can show “paid to the hospital” from the decision alone. Verify actual settlement in Mock Console for that demo run.

Journey evidence · Cashless verification procedure · Retention and authority

One agent, multiple paths

Each branch updates the existing thread and returns to evidence review, a human approval, or a provider check.

Path / trigger	Branch response	Exit evidence / rejoin
Standard reimbursement	Confirm evidence, approve the pack, send once, read decision and match bank credit.	Payer reference, then verified credit.
Missing or contradictory evidence	Flag the wrong patient, dates, bill total or policy period. Request the specific correction; recompute the pack.	Correct evidence clears the blocker; rejoin C1.
User changes intent or authority	Change policy before submission, choose a document version, or withdraw caregiver consent. Recheck approvals and scope.	New user choice or valid authority; rejoin O2/C1.
Provider delay or lost answer	Leave status unconfirmed, look up the client reference, adopt a found receipt; retry only after a negative lookup.	One provider record; rejoin M1.
Repeated renamed request, no acknowledgement	Compare the requested item with prior evidence and dates, ask for acknowledgement, add anything missing and review reply.	Payer accepts response; rejoin waiting.
Partial settlement and short credit	Keep decision disagreement and credit mismatch separate. Draft dispute, record filing, then reconcile revision or top-up.	Matched credit or documented unresolved item; rejoin P1.
Caregiver consent changes; second policy	Pause writes, reconfirm authorized scope, preserve first decision and choose second cover for eligible balance.	Fresh approval and separate second-claim reference.
Post-closure change	A revised decision, late credit or unresolved award obligation reopens the affected part of the file.	Reconciled revision or verified outcome without restarting intake.

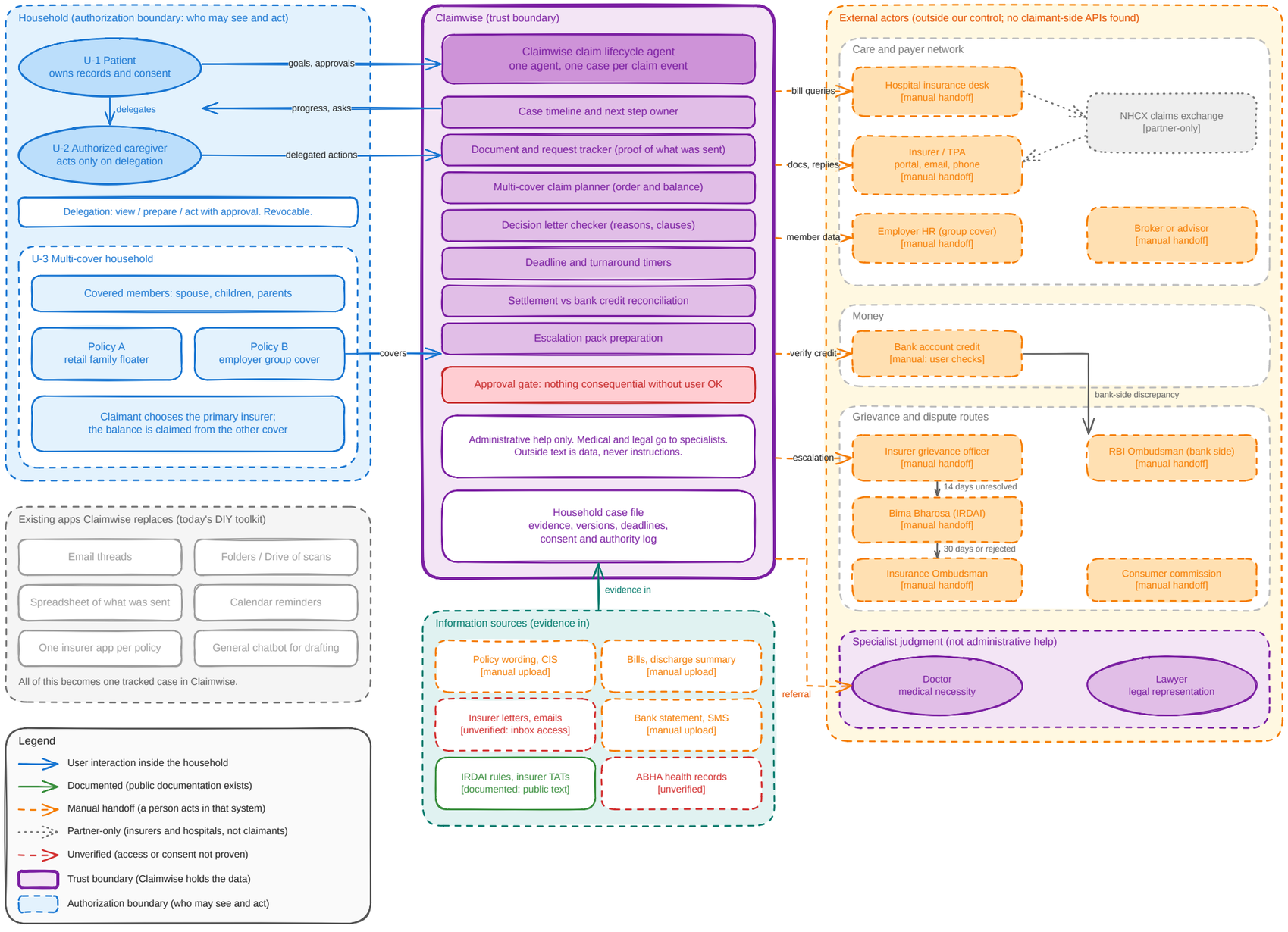
Branch matrix · A–E walkthroughs and branch evidence · Allowed transitions · Research gaps

Product context

Users, provider boundaries, data flows and verified access categories.

G1 Product context graph: users, delegated roles, the agent, existing tools, information sources and external actors

India, health insurance claims. Every integration is labelled documented / partner-only / manual handoff / unverified. A website or app is not proof of API access.



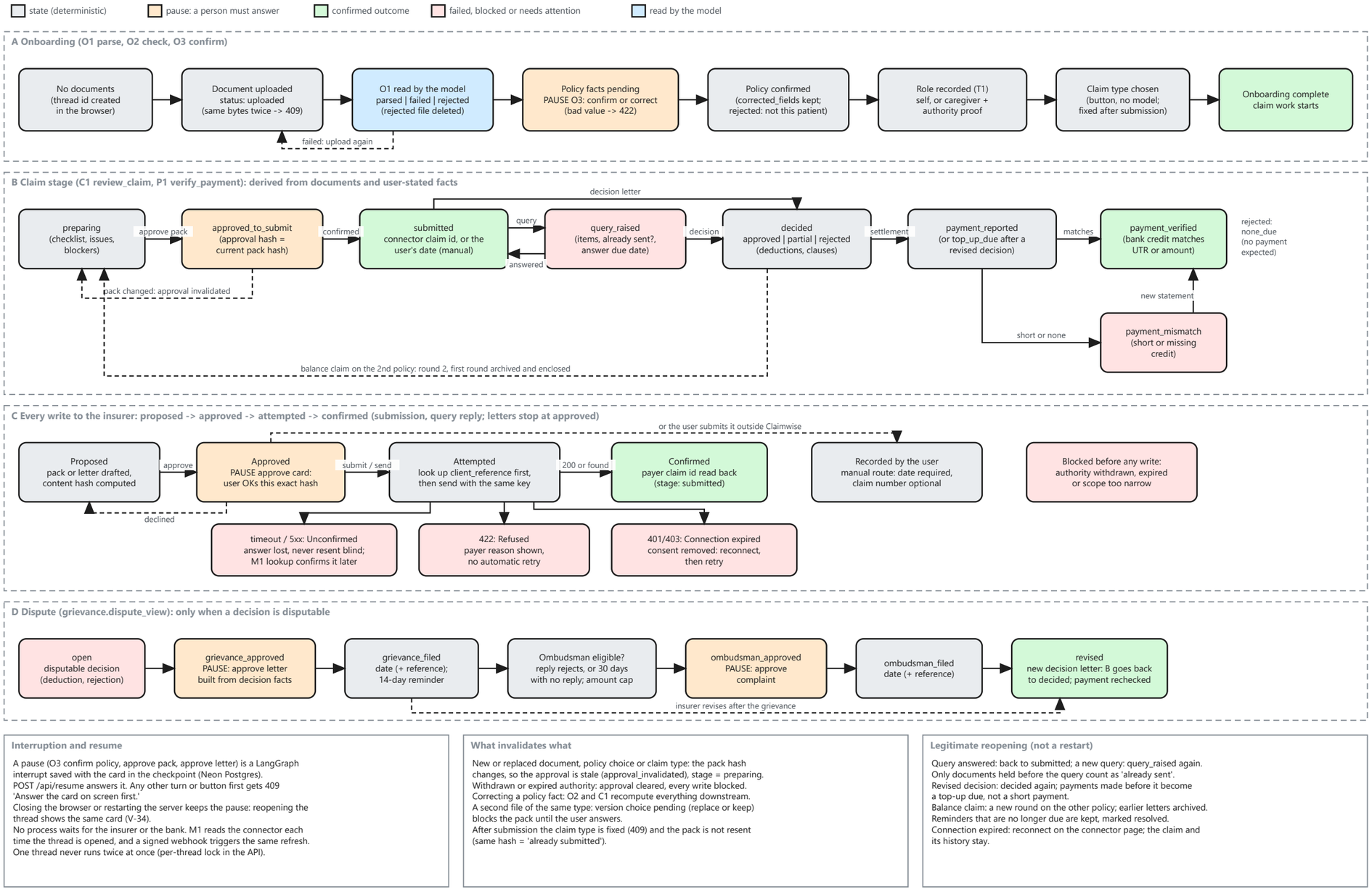
Reading guide · Editable Excalidraw source · Zoomable SVG · Shared legend and function IDs

Lifecycle and state

Human pauses, legal state changes, proof of completion and reopening.

G2 Lifecycle and state graph (BUILT, verified live 2026-11-06, session 037)

Four state machines of one claim thread. Stage and payment status are recomputed from evidence every turn (C1, P1); nothing is marked done by hand. Function ids match G3.



Interruption and resume

A pause (O3 confirm policy, approve pack, approve letter) is a LangGraph interrupt saved with the card in the checkpoint (Neon Postgres).
POST /api/resume answers it. Any other turn or button first gets 409 'Answer the card on screen first.'
Closing the browser or restarting the server keeps the pause: reopening the thread shows the same card (V-34).
No process waits for the insurer or the bank. M1 reads the connector each time the thread is opened, and a signed webhook triggers the same refresh.
One thread never runs twice at once (per-thread lock in the API).

What invalidates what

New or replaced document, policy choice or claim type: the pack hash changes, so the approval is stale (approval_invalidated), stage = preparing.
Withdrawn or expired authority: approval cleared, every write blocked.
Correcting a policy fact: O2 and C1 recompute everything downstream.
A second file of the same type: version choice pending (replace or keep) blocks the pack until the user answers.
After submission the claim type is fixed (409) and the pack is not resent (same hash = 'already submitted').

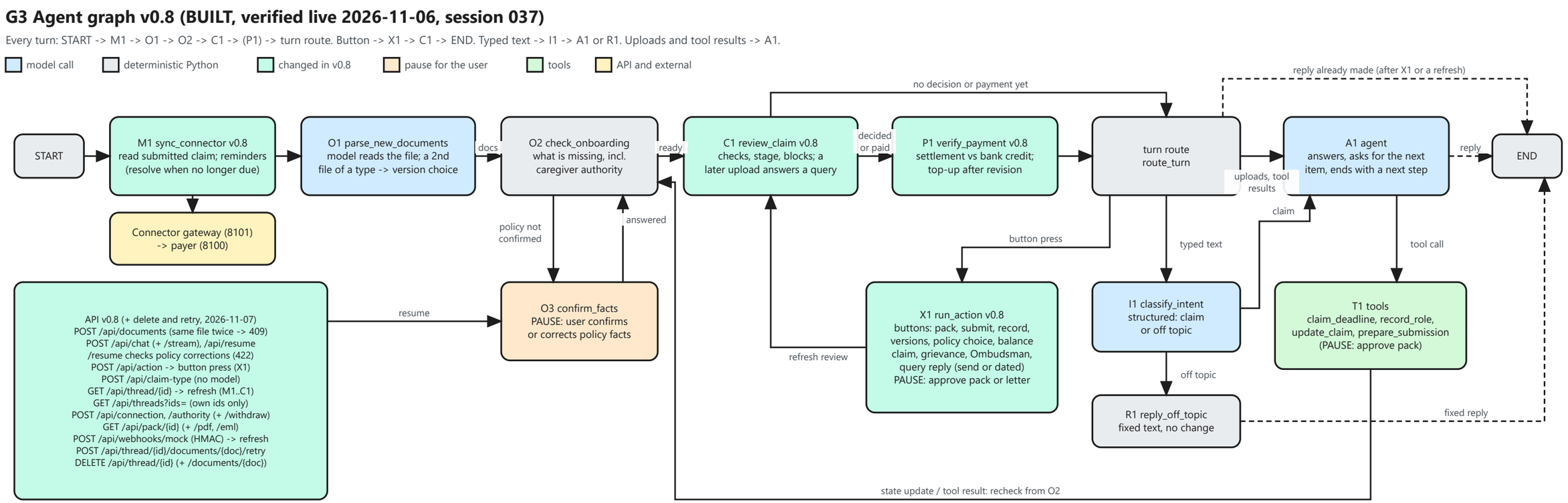
Legitimate reopening (not a restart)

Query answered: back to submitted; a new query: query_raised again.
Only documents held before the query count as 'already sent'.
Revised decision: decided again; payments made before it become a top-up due, not a short payment.
Balance claim: a new round on the other policy; earlier letters archived.
Reminders that are no longer due are kept, marked resolved.
Connection expired: reconnect on the connector page; the claim and its history stay.

Reading guide · Editable Excalidraw source · Zoomable SVG · Shared legend and function IDs

Function execution

One sequential LangGraph turn, with typed data edges, pauses and failure exits.



STATE (checkpoint per thread_id)

messages; documents [{id, doc_type, status, facts, source,
version, uploaded_at}]; policies; user_role; onboarding
claim_info {claim_type, submitted_on, claim_number, round}
submission_approval {hash} (stale when the pack changes)
connection {case_id, status} (case id never leaves the server)
submission {route, status, claim_id, client_reference}
notifications [{id, kind, read, resolved}]; requested_action
authority; audit; dispute; claims_history
Derived each turn: claim (C1) + claim.payment (P1)

C1 / P1 CHECKS (no model)

required documents; names and dates agree; policy period, totals
query: item already sent (only documents held before the query)
decision: per-line deductions, clause not cited
P1: bank credit by UTR or amount; short or missing payment;
a revised decision after payment -> top_up_due
late or overdue flags; reminders (grievance 14 days, authority
expiry); stale reminders resolved
blocks: version choice pending, 2+ covering policies and none
chosen, authority withdrawn, expired or too narrow

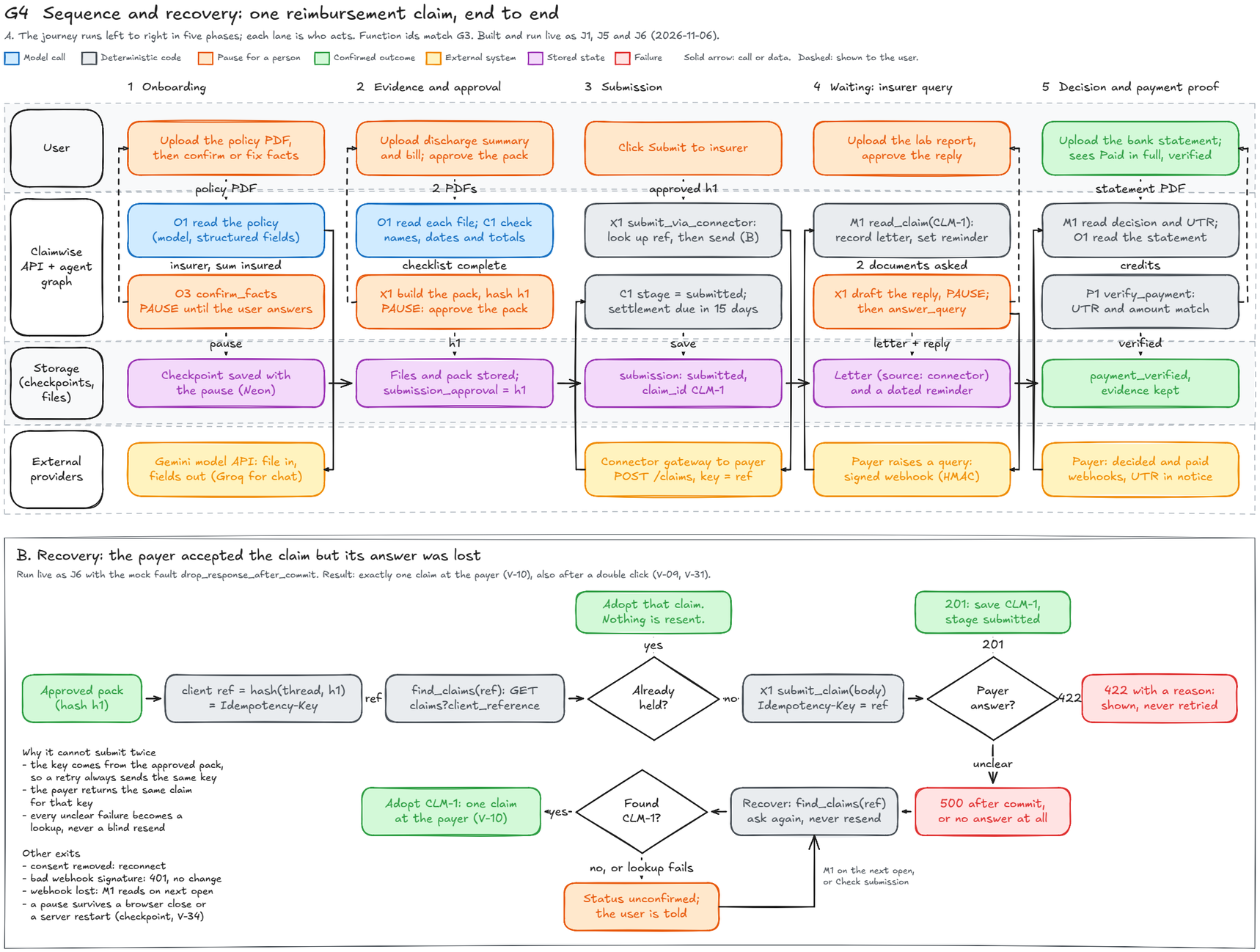
NEW IN v0.8 (session 037, found by the live end-to-end run)

X1: send the documents an insurer query asked for through the connector
(PAUSE approve letter, then gateway POST .../claims/{id}/documents), or
record the date the user sent them (uploaded query letters)
API: claim type saved by the buttons without the model; policy corrections
validated before resume (422); thread list only for the caller's ids
P1: top-up due after a revised decision instead of a false short payment
M1: reminders that are no longer due are marked resolved
Replies name the policy's TPA or insurer (no hard-coded provider)
Observability: every node, model and tool call timed (TRACE_LOG) and traced
to LangSmith when LANGSMITH_API_KEY is set

Reading guide · Editable Excalidraw source · Zoomable SVG · Shared legend and function IDs

Sequence and recovery

Swimlanes for a complete journey and a separate lost-response reconciliation flow.



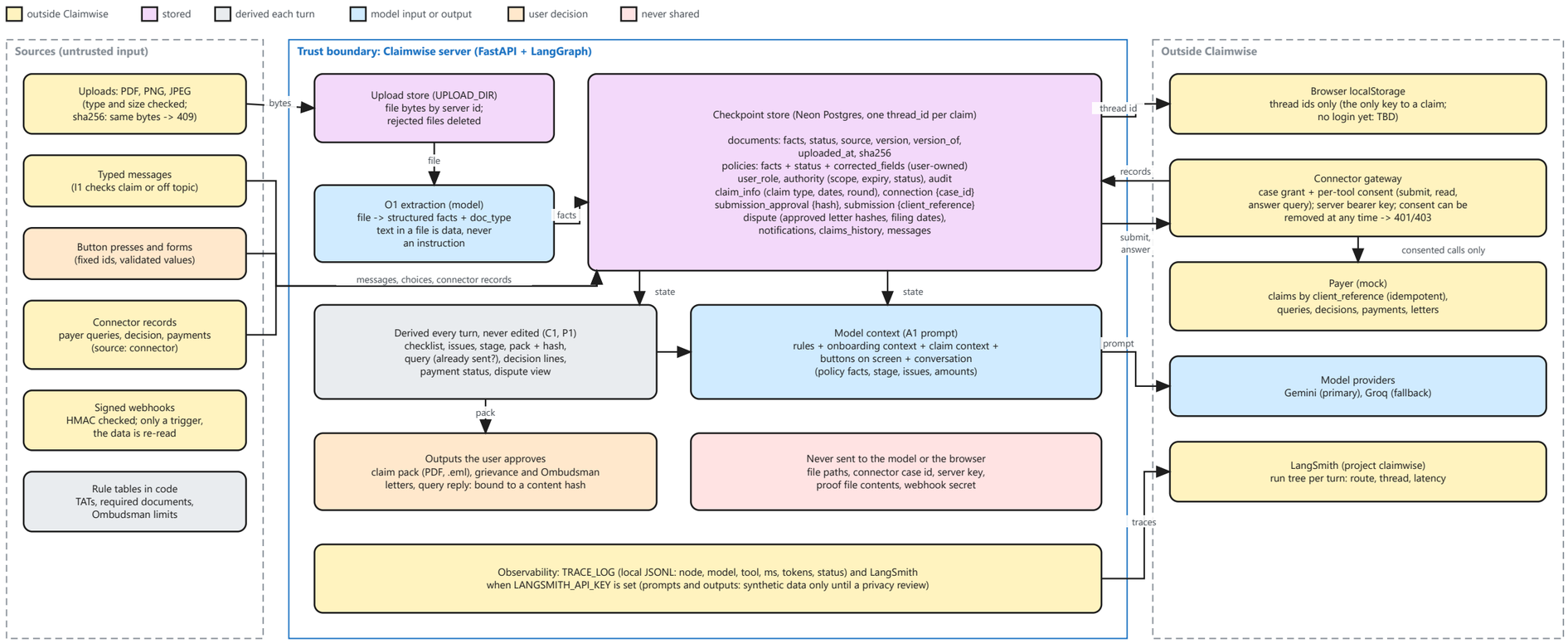
Reading guide · Editable Excalidraw source · Zoomable SVG · Shared legend and function IDs

Data and context

Evidence provenance, storage, versioning and the model's view of the case.

G5 Data and context graph (BUILT prototype, synthetic data only, 2026-11-06; updated 2026-11-08)

Where each fact comes from, who owns it, where it is stored, what the model sees, and how a correction or a new version flows downstream.



Ownership and provenance

Every document keeps its source: an upload (default) or the connector (source: connector, seen time in uploaded_at).
Policy facts: read by the model, owned by the user once confirmed; corrected_fields records which ones the user changed.
Role, claim type, dates and references: stated by the user (buttons or forms), validated at the API.
Authority: recorded by the caregiver with proof; the patient is the grantor; Claimwise records it, it cannot grant it.
Stage, issues, payment and dispute status: derived, never stored as a user decision, so they cannot drift from the evidence.
Audit log: who did what and when (every button action, claim type, authority recorded or withdrawn).
Connector case id and server key stay on the server.

Versions, corrections and approval invalidation

A 2nd file of the same type: version_of set, choice pending; replace -> old one superseded; keep -> both stay in the pack.
A balance claim archives the first round's insurer letters and encloses them; history keeps the first decision.
Correction flow: user corrects a policy fact -> policies updated -> O2 recheck -> C1 recompute -> pack hash changes -> approval stale (approval_invalidated) -> user approves again.
Letters: an approved grievance or Ombudsman letter is bound to its hash; a revised decision starts a new dispute view.
Withdrawn authority clears the pack approval and blocks writes.
Query 'already sent' counts only documents held before the query.

Retention and deletion (prototype status)

Built: rejected uploads are deleted at once; superseded and archived documents are kept as claim evidence.
Built: the thread list only returns ids the browser already holds; the server never lists every claim.
Built: the user can delete one document (purged from current and earlier checkpoints, file removed) or a whole session.
Not built (TBD before real users): user login and tenancy, a retention period, export of a user's data.
LangSmith receives prompts and outputs: keep it to synthetic data, or turn tracing off, until a privacy review approves it.
Secrets (model keys, LangSmith key, connector key, webhook secret) live in backend/.env, never in state or logs.

Context classes: who it belongs to and how far it travels

Personal (one person, in this claim's thread): the patient's medical documents (discharge summary, bills, lab reports), the bank statement, messages; the caregiver's proof of authority.
Household or shared: policy facts that cover several members (family floater, employer cover), the authority the patient grants a caregiver, the choice of which policy to claim first.
Built: held in each claim's thread. Proposed: a household profile shared by the family's claims.
Task-specific (this claim only): claim type, pack and approval hashes, submission and client reference, queries, decision, payment, dispute, reminders, audit.
Reusable across cycles: rule tables in code (TATs, required documents, Ombudsman limits) and the extraction schema. Proposed, not built: reuse confirmed policy facts and an unexpired authority for the next claim. Today each claim starts a new thread and nothing is carried over automatically.

Reading guide · Editable Excalidraw source · Zoomable SVG · Shared legend and function IDs

Execution and responsibility

React collects choices. FastAPI validates input. LangGraph schedules the turn and persists the state. Domain calculations run in Python.

IDs	Responsibility and handoff
M1 · sync_connector	Deterministic status lookup and reminders. Read only under active grants. Adopt a confirmed receipt or mark connection expired; hand evidence to O1/C1.
O1 → O2 → O3	Gemini extracts typed document facts. O2 determines missing items. O3 pauses for human policy confirmation or corrections, then rechecks. Failed extraction remains retryable.
C1 → P1	Deterministic document checks, pack hash, stage and settlement review. Compare bank evidence when applicable. A discrepancy becomes a blocker or next action.
X1	Run an offered action with exact inputs. Validate authority and approval, then export, write through the mock gateway or hand off.
I1 / R1 · A1 / T1	Classify typed intent, return a fixed off-topic reply or draft a grounded reply. Tools update typed state. The model cannot authorize a provider write.

Why this stack

React and Vite provide one case workspace. FastAPI and Pydantic type the HTTP and extraction boundaries. LangGraph supplies checkpoints and interruptions. Neon retains turns across restarts. FastAPI and SQLite isolate the mock external world. Gemini handles documents; Groq is a text fallback.

Retrieval boundary. Current evidence comes from user uploads, scoped mock observations and rule tables. The prototype has no vector retrieval. Unconfirmed facts go back to a human, while sums, dates, deadlines, eligibility gates and writes stay deterministic.

Typed function contracts, normal and failure examples · Models and retrieval · Stack rationale · Executable graph

Interfaces and external access

Implemented internal endpoints use the /api prefix. Live external access is evaluated separately from the mock proof.

Operation or integration	Contract, approval and fallback
POST /documents · POST /resume	Multipart thread and file, or approved and corrections. Validates type, size and thread; returns the next turn or pending card.
POST /action · POST /chat/stream	Offered action and typed fields, or thread and text over SSE. Recheck pending state after an interrupted request.
GET /threads · GET /thread/{id}	Supplied browser thread IDs and current snapshot with history. Demo access model needs authenticated tenancy for live use.
Retry and delete	Retry an incomplete document by ID or re-upload identical failed bytes. Document delete removes that file and historical copies; session delete clears its checkpoints and uploads.
Mock gateway / payer	Server bearer and active scoped grant. Submit requires current approval and stable idempotency key. Lookup client reference before replay. Manual pack export if unavailable.
Insurer, TPA, hospital, bank	Live access unverified or partner-only. Obtain documents or use the authorized human route. NHCX profiles are documented; Claimwise access is unverified.
Grievance and Ombudsman	Public filing channels documented. Claimwise drafts and records evidence; an eligible human files.

Payload example · Policy confirmation

```
{
  "thread_id": "demo_claim_001",
  "approved": true,
  "corrections": {
    "policy_number": "DEMO-POLICY-001",
    "period_end": "2027-03-31"
  }
}
```

On success, the graph resumes the pending card and returns a refreshed case. Missing pending approval or invalid corrections produce an explicit error.

Methods, access, limits and fallbacks · Schemas · Example request · Example response · Error catalogue · Current API

Context, permissions, and recovery

Patient facts, delegated authority and claim evidence travel with the case and have distinct owners and lifetimes.

Context class	Current handling and design boundary
Personal	Patient and representative fields are case-scoped. Proof and expiry are checked. Identity verification and production tenant isolation remain gates.
Household or shared	Confirmed policy members and multiple covers support policy choice and a separate balance round. Sharing authority must be scoped and revocable.
Task-specific	Files, derived facts, approvals, receipts, queries, decisions, disputes and payment evidence stay in the claim thread. Correction recomputes pack hash and can invalidate approval.
Reusable	Rules, document types, action patterns and deterministic check functions are reused across cases. Future reuse of personal or household evidence requires approved retention and access policy.

Execution and crash recovery

A process-local lock serializes one thread. A turn is sequential and separate threads can proceed independently. Human waits persist as interrupts. Reopen and signed mock events trigger checks; there is no durable background scheduler. Neon and the original file store have separate lifetimes.

Authority and deletion

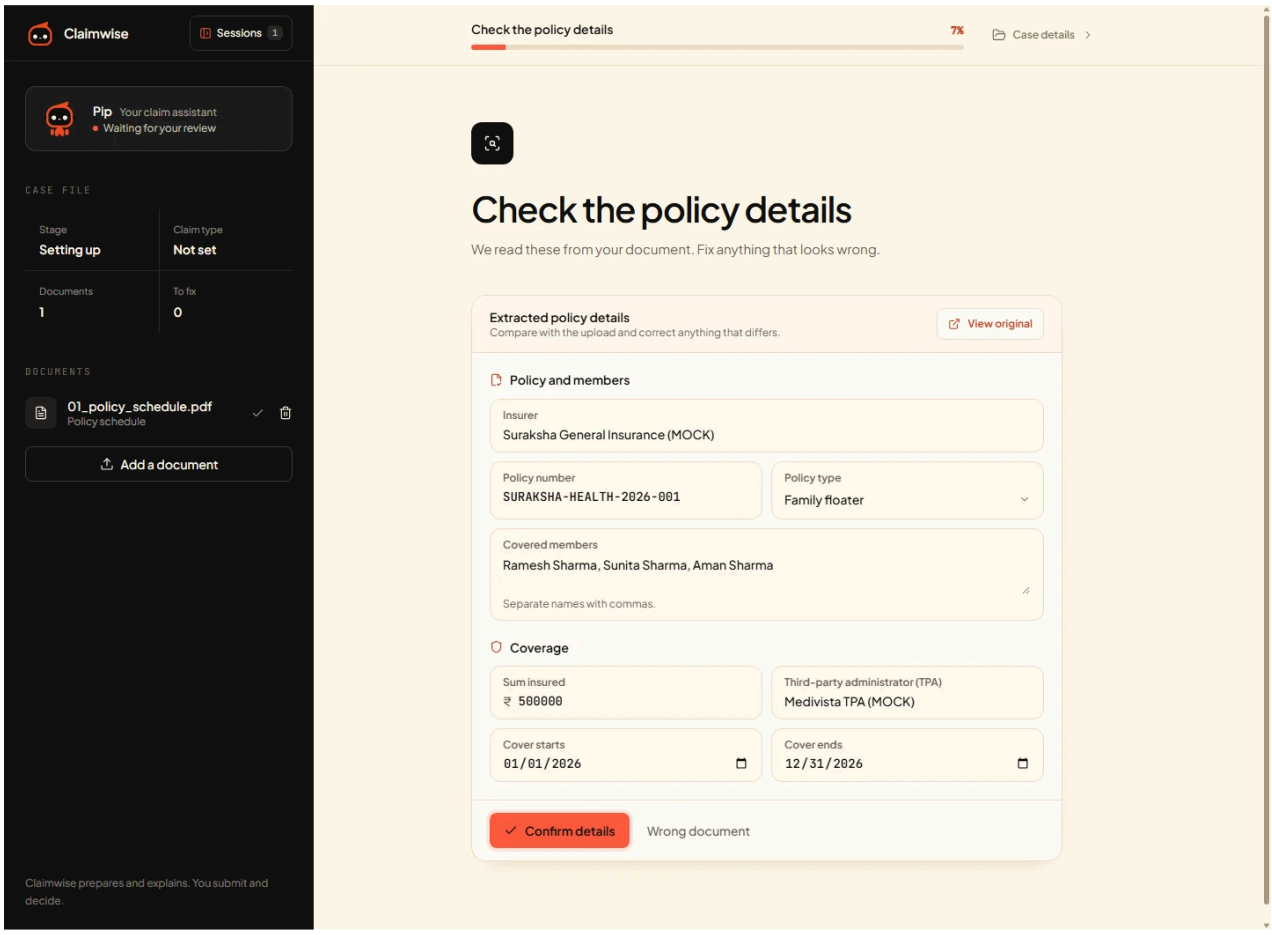
Use a scoped mock grant and server-side key, verify the current pack and authority, record audit evidence, and stop affected writes on revocation. Imported text cannot approve an action. Local session and upload deletion are implemented. Production retention, export, authentication and tenant isolation remain open.

Diagram currency. G3 includes the current retry and deletion routes. G5 distinguishes personal, household, task-specific and reusable context and identifies the demo deletion boundary.

Execution and memory · Permissions and data · Deletion and retry source · State schema

The user sees what needs a decision

The Dossier shows the current step, progress, case evidence and Pip's activity. The policy card below came from a real upload of a repository fixture.



Screen	User control and evidence
Onboarding and goal	Upload policy, correct extracted facts, identify patient and scope, select route or cover.
Progress and needs-input	Read the next action, supporting documents, blockers, provider notices and outstanding evidence.
Comparison and approval	Inspect policy facts, deductions or payment matches. Review the current pack or letter before accepting a consequential step.
Outcome and correction	See evidence for submission, decision and payment separately. Upload corrected evidence or reopen the affected decision.

All screen sketches · Short demo sequence and full walkthrough · Synthetic fixtures

From first input to proof

These journeys summarize the recorded synthetic run. Each handoff separates user action, function, state and outcome.

Journey	Inputs, functions and state transitions	External action and outcome evidence
A · Standard reimbursement	Policy upload → O1/O3 confirm → role and route → discharge/bill C1 checks → X1 pack approval → submitted → decided → P1 verification.	Gateway submits once. Mock payer provides a claim number and reports payment. A bank statement match by UTR proves reimbursement. J1.
B · Caregiver and dispute	Consent proof → authority active → revocation clears approval → re-authorize and submit → partial decision → grievance and Ombudsman drafts → revised decision and top-up due.	Mock payer checks representative. User records filing after manual handoff. Revised decision proves reopening. J2 reached top-up due. Matching the later top-up is a specification that has not been run.
C · Failure and follow-up	Renamed request → C1 flags repeat and collects missing evidence. Lost answer → X1/M1 mark unconfirmed and look up the client reference. Without connection, use manual filing and record the date.	One payer record after reconciliation. On the manual branch, the user handles the request and the short bank credit remains a payment mismatch. J5–J7.

Additional branches

J3 exercises a household balance claim on the second policy after the first partial decision. J4 shows a representative refused by the mock payer even after the app records their authority.

Full input, function, state, external action and evidence traces · Recorded checks · Current walkthrough with fixtures

Evidence, including the open checks

6 November 2026 baseline. The repository records 124 of 127 live checks passing across 37 specifications and 13 journeys, plus 52 backend tests for that run.

Spec	Expected outcome
V-01 / V-03	Policy waits for confirmation; missing context keeps onboarding open.
V-05 / V-07	Evidence inconsistency raises a blocker; changed pack invalidates approval.
V-09 / V-10	Duplicate submissions create one claim; lost response is reconciled before replay.
V-11 / V-12	Signed webhook refreshes; forged event is refused. Renamed duplicate request is detected.
V-13 / V-14	Reply evidence requires approval; decision deductions are compared and missing clauses flagged. Three live checks remained open.
V-15 / V-16	UTR match verifies reimbursement; short or missing credit stays unresolved.
V-18 / V-20	Repeated file is recognized, corrected file is versioned, and consent withdrawal blocks writes.
V-22 / V-25	Second cover sees the eligible balance; revision reopens the affected check.
V-26 / V-27	Malicious document text, chat or file name cannot authorize an action.
V-30 / V-35	Cross-thread original is refused and list returns only supplied thread IDs. Demo boundary only.
V-31 / V-34	Concurrent same-thread requests are serialized and Postgres retains pauses and state after restart.

Open results. One V-13 live recheck remained after a code fix; two V-14 extraction checks hit the provider quota. This submission did not run application tests. The screenshot capture exercised policy upload only.

Open results and all 37 specs · Historical measurements · Reproduction instructions

A supervised path to a pilot

The proposed release keeps consequential live actions with authorized humans until identity, provider access and recovery are proven.

Phase	Deliverable or gate
Foundation and evidence	Workspace, checkpointing, document validation and pack export.
Guided turns and monitoring	State-based next actions, approval pauses, connected mock and manual handoffs.
Domain extensions	Caregiver authority, document versions, second cover, grievance and payment reopening.
Packaging	Vercel frontend, Render API and mock services, separate connector gateway, and Neon when configured. Verify deployed origins and storage.
Pilot gate	Start with supervised synthetic cases. Real-user pilot requires approved identity, permissions, access agreements, retention and safe operations.

Cost and measurement

Estimate fixed hosting plus database and file storage, model input/output and retries, and human review and support. The worked standard reimbursement example estimates USD 0.00810 in model calls using recorded aggregate token means and a stated call mix. One extra Groq call adds about USD 0.00026. Hosting, storage, egress and review time are separate. Historical timing spans inform the measurement plan.

Success, stop, and rollback

Measure evidence-backed completion, first-pass pack acceptance, and unresolved requests. Track isolation incidents, unapproved writes, duplicates and ambiguous outcomes. Pause writes and revoke affected access on an incident. Reconcile provider receipts before replay. Roll back the deployment independently of actions a provider already accepted.

Implementation milestones · Worked cost and pilot gates · Measurement evidence · Hosting and demo boundaries · Research references